

Mockito Hands-On Exercises

Exercise 1: Mocking and Stubbing

Scenario:

You need to test a service that depends on an external API. Use Mockito to mock the external API and stub its methods.

Steps:

1. Create a mock object for the external API.
2. Stub the methods to return predefined values.
3. Write a test case that uses the mock object.

Solution Code:

```
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test; import
org.mockito.Mockito;

public class MyServiceTest {
    @Test
    public void testExternalApi() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");
        MyService service = new MyService(mockApi);
        String result = service.fetchData();    assertEquals("Mock
        Data", result);
    }
}
```

```
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an age
nt to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.514 s -- in com.example.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.206 s
[INFO] Finished at: 2026-06-21T16:13:18+05:30
```

Exercise 2: Verifying Interactions

Scenario:

You need to ensure that a method is called with specific arguments.

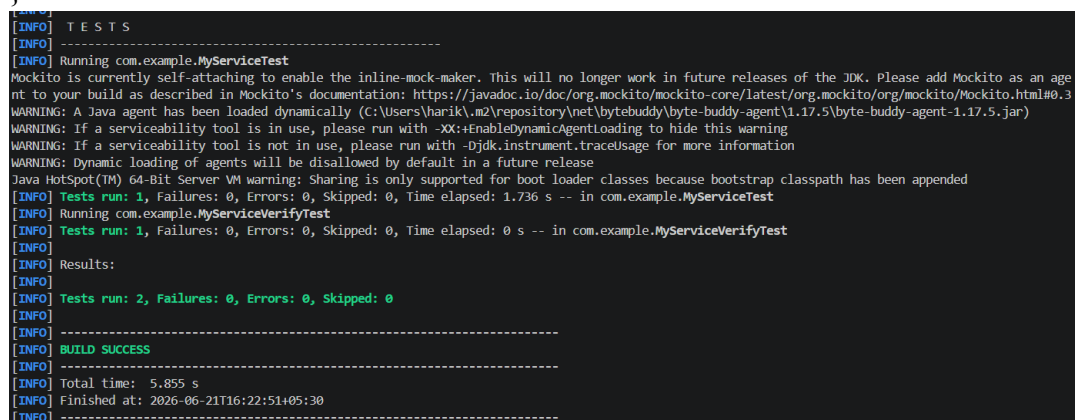
Steps:

1. Create a mock object.
2. Call the method with specific arguments.
3. Verify the interaction.

Solution Code:

```
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test; import
org.mockito.Mockito;
```

```
public class MyServiceTest {
    @Test
    public void testVerifyInteraction() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        MyService service = new MyService(mockApi);
        service.fetchData();    verify(mockApi).getData();
    }
}
```



```
[INFO] TESTS
[INFO] -----
[INFO] Running com.example.MyServiceTest
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to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
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WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.736 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVerifyTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 5.855 s
[INFO] Finished at: 2026-06-21T16:22:51+05:30
[INFO] -----
```

Exercise 3: Argument Matching

Scenario:

You need to verify that a method is called with specific arguments.

Steps:

1. Create a mock object.
2. Call the method with specific arguments.
3. Use argument matchers to verify the interaction.

```

[INFO] TESTS
[INFO] Running com.example.MyServiceArgumentTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.607 s -- in com.example.MyServiceArgumentTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.MyServiceVerifyTest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 5.722 s
[INFO] Finished at: 2026-06-21T16:29:17+05:30

```

Exercise 4: Handling Void Methods

Scenario:

You need to test a void method that performs some action.

Steps:

1. Create a mock object.
2. Stub the void method.
3. Verify the interaction.

```

[INFO] TESTS
[INFO] Running com.example.MyServiceArgumentTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.455 s -- in com.example.MyServiceArgumentTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s -- in com.example.MyServiceVerifyTest
[INFO] Running com.example.MyServiceVoidTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVoidTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 5.355 s
[INFO] Finished at: 2026-06-21T16:33:50+05:30

```

Exercise 5: Mocking and Stubbing with Multiple Returns

Scenario:

You need to test a service that depends on an external API with multiple return values.

Steps:

1. Create a mock object for the external API.

2. Stub the methods to return different values on consecutive calls.
3. Write a test case that uses the mock object.

```
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceArgumentTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent
to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.703 s -- in com.example.MyServiceArgumentTest
[INFO] Running com.example.MyServiceMultipleReturnsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceMultipleReturnsTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.MyServiceVerifyTest
[INFO] Running com.example.MyServiceVoidTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -- in com.example.MyServiceVoidTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.768 s
[INFO] Finished at: 2026-06-21T16:36:43+05:30
```

Exercise 6: Verifying Interaction Order

Scenario:

You need to ensure that methods are called in a specific order.

Steps:

1. Create a mock object.
2. Call the methods in a specific order.
3. Verify the interaction order.

```
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceArgumentTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent
to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.613 s -- in com.example.MyServiceArgumentTest
[INFO] Running com.example.MyServiceMultipleReturnsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceMultipleReturnsTest
[INFO] Running com.example.MyServiceOrderTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceOrderTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVerifyTest
[INFO] Running com.example.MyServiceVoidTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVoidTest
[INFO] Results:
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.691 s
[INFO] Finished at: 2026-06-21T16:39:24+05:30
```

Exercise 7: Handling Void Methods with Exceptions

Scenario:

You need to test a void method that throws an exception.

Steps:

1. Create a mock object.
2. Stub the void method to throw an exception.
3. Verify the interaction.

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceArgumentTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\harik\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.5\byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.487 s -- in com.example.MyServiceArgumentTest
[INFO] Running com.example.MyServiceExceptionTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceExceptionTest
[INFO] Running com.example.MyServiceMultipleReturnsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s -- in com.example.MyServiceMultipleReturnsTest
[INFO] Running com.example.MyServiceOrderTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceOrderTest
[INFO] Running com.example.MyServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceTest
[INFO] Running com.example.MyServiceVerifyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVerifyTest
[INFO] Running com.example.MyServiceVoidTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.MyServiceVoidTest
[INFO] Results:
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.694 s
[INFO] Finished at: 2026-06-21T16:41:55+05:30
```